CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

November 2012

EC 201 Digital Electronics & Logic Design

Date: 27.11.2012, Tuesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q - 1 **(a)** Perform the subtraction $(5720)_{10}$ - $(1230)_{10}$ using 9's complement and 10's complement. **[03]**

(b) Obtain 5's and 6's complement of $(543210)_6$. **[02]**

(c) Which code is used in the K - map? **[01]**

(d) What do you mean by don't care? **[01]**

Q - 2 **(a)** Give three possible ways to express the function: **[07]**

F = A'B'D'+AB'CD'+A'BD+ABC'D

with eight of less literals.

(b) Design a combinational circuit that converts a decimal digit from the BCD code to Excess-3 code. **[07]**

OR

Q - 2 **(a)** Simplify the following Boolean function by means of the tabulation method **[07]**

$F(A,B,C,D,E,F,G) = \sum (20,28,52,60)$

(b) Design a combinational circuit that converts a decimal digit from the 2,4,2,1 code to 8,4,-2,-1 code. **[07]**

Q - 3 **(a)** Implement a full-subtractor with two half subtractors and an OR gate. **[04]**

(b) Explain carry propagation in detail. **[05]**

(c) Construct a 6×64 Decoder with 3×8 Decoders. **[05]**

OR

Q - 3 **(a)** Implement Full adder combinational circuit. **[04]**

(b) Implement function **[05]**

$$F(A,B,C,D) = \sum(2,4,6,7,9,13,15)$$

with ROM.

(c) Write down the design procedure steps for designing any combinational circuit. **[05]**

SECTION – II

Q - 4 (a) Reduce the following Boolean expressions to the required number of literals [05]

(i) ABC+A'B'C+A'BC+ABC'+A'B'C' to five literals

(ii) BC+AC'+AB+BCD to four literals

(b) Demonstrate by means of truth table the validity of Demorgan's theorems [02]

Q - 5 (a) What do you mean by sequential logic? Explain Edge triggered flipflop. [07]

(b) Design a counter with the following sequence: 0, 4, 2, 1, 6 and repeat. Use JK flip-flop. [07]

OR

Q - 5 (a) What do you mean by master-slave flipflop? Explain clocked master-slave JK flipflop. [07]

(b) Design a counter with the following sequence:0, 1, 3, 2, 6, 4, 5, 7 and repeat. Use RS flipflops.. [07]

Q - 6 (a) Design 4 bit Binary Ripple Counter. [07]

(b) Design 4-bit bidirectional shift register with parallel load. [07]

OR

Q - 6 (a) Design 4 bit binary Synchronous Counter. [07]

(b) Explain Johnson Counter. [07]

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